

Chromosomal inversion at the *DG1* promoter drives double-grain spikelets and enhances grain yield in sorghum

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The phenomenon of multiple-grain spikelets is frequently observed in gramineous crops. In the case of dual-floret spikelets, the upper fertile floret develops normally to form a single grain, while the lower sterile floret undergoes abortion. Here we elucidate the role of *Double-Grain1* (*DG1*), a gene encoding a homeobox-domain-containing protein, in regulating the lower floret meristem activity and double-grain spikelet trait in sorghum. A 35.7-kb paracentric inversion in the *DG1* promoter region leads to increased *DG1* expression, probably by reducing repressive histone modifications. This increase in *DG1* expression transforms the degenerated lower floret into a fertile one. The use of the superior *DG1* allele results in an increase of approximately 40.7% to 46.1% in grain number per panicle and a 10.1% to 14.3% increase in overall grain yield. Our findings shed light on the sorghum double-grain spikelet characteristic, offering valuable insights for high-yield breeding designs in cereals.

A typical cereal inflorescence comprises a fixed or indeterminate number of spikelets along the main axis, each containing one or multiple florets, determined by the timing of spikelet meristem and floret meristem termination. The spikelet, a higher-order floral organ, self-repeats along the axis and usually includes two or more florets. In a typical dual-floret spikelet, such as sorghum, the upper floret is consistently fertile, giving rise to a single grain, while the lower floret remains sterile. Often, the sterile floret initiates but undergoes early developmental abortion, ultimately degenerating into a lemma- or glume-like structure¹. Although single-grain spikelets are prevalent in crops, double-grain spikelet morphology (possibly resulting from human selection) exists in a limited number of germplasm accessions. The double-grain spikelet trait contributes to an increased grain number per panicle, thereby enhancing the final grain yield².

Sorghum, the fifth-largest cereal crop globally, is renowned for its high photosynthetic efficiency and resilience to various abiotic stresses, including drought, salinity, alkalinity and waterlogging^{3–6}. Its grain yield is influenced by three primary components: panicle number, grain number per panicle and grain weight⁷. Earlier research has established grain number as the principal determinant of grain yield in sorghum, explaining 52% of the variation in yield^{8–10}. Sorghum displays a fertile sessile spikelet (SS), which produces a single grain, directly attached to the tertiary branch. This is accompanied by a sterile pedicellate spikelet (PS) positioned closely adjacent to the SS through a pedicel¹. The number of PSs and SSs greatly affects sorghum grain number and yield. Previous investigations have identified three genes—*Multiseeded1* (*MSD1*), *MSD2* and *MSD3*—as pivotal in regulating the fertility of PSs in sorghum. *MSD1* encodes a TCP family transcription

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factor¹¹, *MSD2* encodes a lipoxygenase¹² and *MSD3* encodes an ω -3 fatty acid desaturase catalysing the conversion of linoleic acid to linolenic acid—a substrate for jasmonic acid synthesis¹³. Mutations in all three genes lead to a reduction in jasmonic acid content, resulting in increased PS fertility and a higher number of grains¹⁰. However, the incomplete development and ultimate abortion of the lower sterile floret in the SS remains unclear. As early as approximately a century ago, scholars reported the occurrence of double-grain sorghum¹⁴.

The floral meristem plays a crucial role in controlling the architecture of plant floret organs, influenced by a conserved negative feedback loop in the *CLAVATA* (*CLV*)–*WUSCHEL* (*WUS*) signalling pathway^{15,16}. The homeobox-containing *WUS* gene is a key regulator in maintaining meristem cell activity, and alterations in its expression level result in abnormal floret development. In *Arabidopsis*, the *wus* mutant exhibits severe defects in stamen and pistil development, characterized by the presence of only one stamen¹⁷. Conversely, ectopic expression of *WUS* leads to an enlargement of the floral meristem, resulting in an increased number of stamens and carpels^{18,19}. In rice, the floral organ number genes *FON1* and *FON2/4*, homologous to *CLV1* and *CLV3* in *Arabidopsis*, negatively regulate floral meristem activity^{20–22}. Mutations in rice *FON1* or *FON2* lead to an enlargement of the floral meristem and an increased number of stamens and pistils. The rice *Twin-grain 1* (*tg1*) gene, allelic to *FON2/FON4*, enhances glume opening and stigma exertion, ultimately increasing rice hybrid seed production². In maize, *ZmWUS1* regulates inflorescence meristem (IM) initiation and maintenance. The *Bif3* mutant, which contains a tandem duplication of *ZmWUS1* and shows enhanced transcription levels due to a new promoter, exhibits an enlarged IM size but a reduced number of spikelet meristems in the tassel and ear²³.

In the realm of epigenetic regulation, DNA methylation, histone modifications, non-coding RNAs and other mechanisms bestow heritable alterations in gene expression without modifying the underlying DNA sequence²⁴. Histone H3 lysine 27 trimethylation (H3K27me3) and histone H3 lysine 9 dimethylation (H3K9me2) represent two prevalent and extensively characterized repressive histone modifications in plants²⁵. Several investigations indicate that histone methylation plays a role in the transcriptional inhibition of plant growth and environmental response. The Jumonji-C domain-containing protein JMJ30 directly interacts with chromatin at *FLOWERING LOCUS C*, targeting the repressive mark H3K27me3 and thereby preventing premature flowering due to high-temperature exposure²⁶. The insertion of a solo-long terminal repeat (LTR) led to increased DNA methylation levels and the accumulation of H3K9me2, collectively leading to the downregulation of *GY3* expression and subsequently promoting high grain yield in rice²⁷. In cotton, the elimination of H3K27me3 from the promoters of jasmonate-related genes resulted in elevated expression levels, which helped sustain male fertility under high-temperature conditions²⁸. However, the repressive histone methylation associated with altered gene expression in sorghum has scarcely been studied.

In this study, we present a sorghum cultivar, NSL360516, showing the distinctive double-grain trait within the SS. Employing positional cloning, we identified a paracentric inversion at the *Double-Grain 1* (*DGI*) locus, which encodes a homeobox-domain transcription factor. This inversion led to reduced accumulation of repressive histone modifications H3K9me2 and H3K27me3, resulting in increased expression associated with the restoration of normal development in the lower sterile floret. The natural superior allele of *DGI* holds substantial promise for applications aimed at enhancing grain number and overall yield in further sorghum breeding.

Results

Map-based cloning of *DGI* locus

The widely common single-grain spikelet and the less common double-grain spikelet were both observed in sorghum varieties (Fig. 1a). The mature double-grain spikelet exhibits two types: an effective

double-grain spikelet (EDGS) bearing two sizable grains, and an ineffective double-grain spikelet (IDGS) producing a single normal grain, with an additional underdeveloped grain transitioning into a trace (Extended Data Fig. 1a). Hereafter, the term ‘double-grain’ essentially refers to EDGSs. Evaluation of 571 sorghum accessions at the mature stage revealed only two inbred lines producing double grains in the SS. The cultivar NSL360516 exhibited double grains in 52.6% of panicles, a higher percentage than the other cultivar, F.R. Miller. To investigate the genetic basis of the double-grain trait, both double-grain accessions were crossed with three single-grain accessions, SC855, KS115 and BTx399, to construct independent segregated populations (Extended Data Fig. 1b). All F₁ plants exhibited the double-grain trait with similar percentages to their double-grain parental lines (Extended Data Fig. 1c,d). In the three F₂ populations derived from NSL360516 and different single-grain lines, the segregation ratio of double-grain to single-grain plants conformed to a 3:1 ratio ($\chi^2_{0.05} = 0.04, 0.77, 1.22$) (Extended Data Fig. 1e), indicating that the double-grain trait is controlled by a single dominant locus, named *DGI*.

By developing 410 simple sequence repeat (SSR) markers in the F₂ population from the crosses between NSL360516 and three single-grain accessions, we initially mapped the *DGI* locus to the end of chromosome 6, flanked by markers *SSR6-46* and *SSR6-66*. To further fine-map *DGI*, we developed enlarged F₃ and F₄ populations derived from the cross between NSL360516 and BTx399, totalling 2,186 individuals. Four additional single nucleotide polymorphism (SNP) markers and one SSR marker were designed within the *DGI* region to genotype all individuals. Seventeen recombinants with five different recombination types were screened, ultimately narrowing down the *DGI* interval to a 49.5-kb region flanked by markers *SNP6-46* and *SNP6-43* (Fig. 1b). This region contains five annotated genes: *Sobic.006G254900*, *Sobic.006G255000*, *Sobic.006G255100*, *Sobic.006G255200* and *Sobic.006G255300*, referred to as *ORF1*, *ORF2*, *ORF3*, *ORF4* and *ORF5*, respectively (Fig. 1c).

To identify which candidate gene contributes to the double-grain trait of NSL360516, we sequenced all five genes between the parental lines and found no variations in the coding sequence region. The focus points were then shifted to possible variations in the promoter region. Sequence analysis of the *ORF2*, *ORF3* and *ORF4* promoters showed no obvious differences between the two parental lines. However, when primers were designed for the predicted -2-kb products of *ORF1* and *ORF5*, no PCR products were obtained in several attempts for NSL360516, probably due to a large DNA fragment insertion, deletion or inversion. We subsequently identified a chromosomal paracentric inversion spanning 35.7 kb across the 5′ terminus of *ORF1* and the 5′ terminus of *ORF5* (Fig. 1c). This inversion was confirmed by the specific primers F₁, R₁, F₂ and R₂. Predicted products of 2.3 kb and 2.5 kb were detected only in BTx399 by F₁ + R₁ and F₂ + R₂, respectively, while predicted 2.0-kb and 2.8-kb products were detected only in NSL360516 by F₁ + F₂ and R₁ + R₂, respectively (Fig. 1c,d). Notably, this inversion directly altered the promoter region sequence of both *ORF1* and *ORF5*, despite the lack of impact on the other three annotated genes. Additionally, sequence analysis within the total inverted segment revealed a 7.4-kb fragment insertion at the edge of the *DGI* locus in NSL360516. This insertion may not affect the gene function of the *ORFs* due to the long-term physical distance from them.

Elevated *ORF1* expression induces double-grain spikelets

In the mapping process, we developed a pair of near-isogenic lines (NILs) at the *DGI* locus through self-crossing and a marker-assisted selection system (Extended Data Fig. 2a). In the F₆ generation, a plant with a heterozygous *DGI* region was selected and self-crossed into the F₇ population. The homozygous dominant and recessive *DGI* alleles were genotyped using flanking markers, resulting in NIL-*dg1* (harbouring the BTx399 allele and exhibiting a single-grain trait) and NIL-*DGI* (carrying the NSL360516 allele and exhibiting a double-grain trait). Less

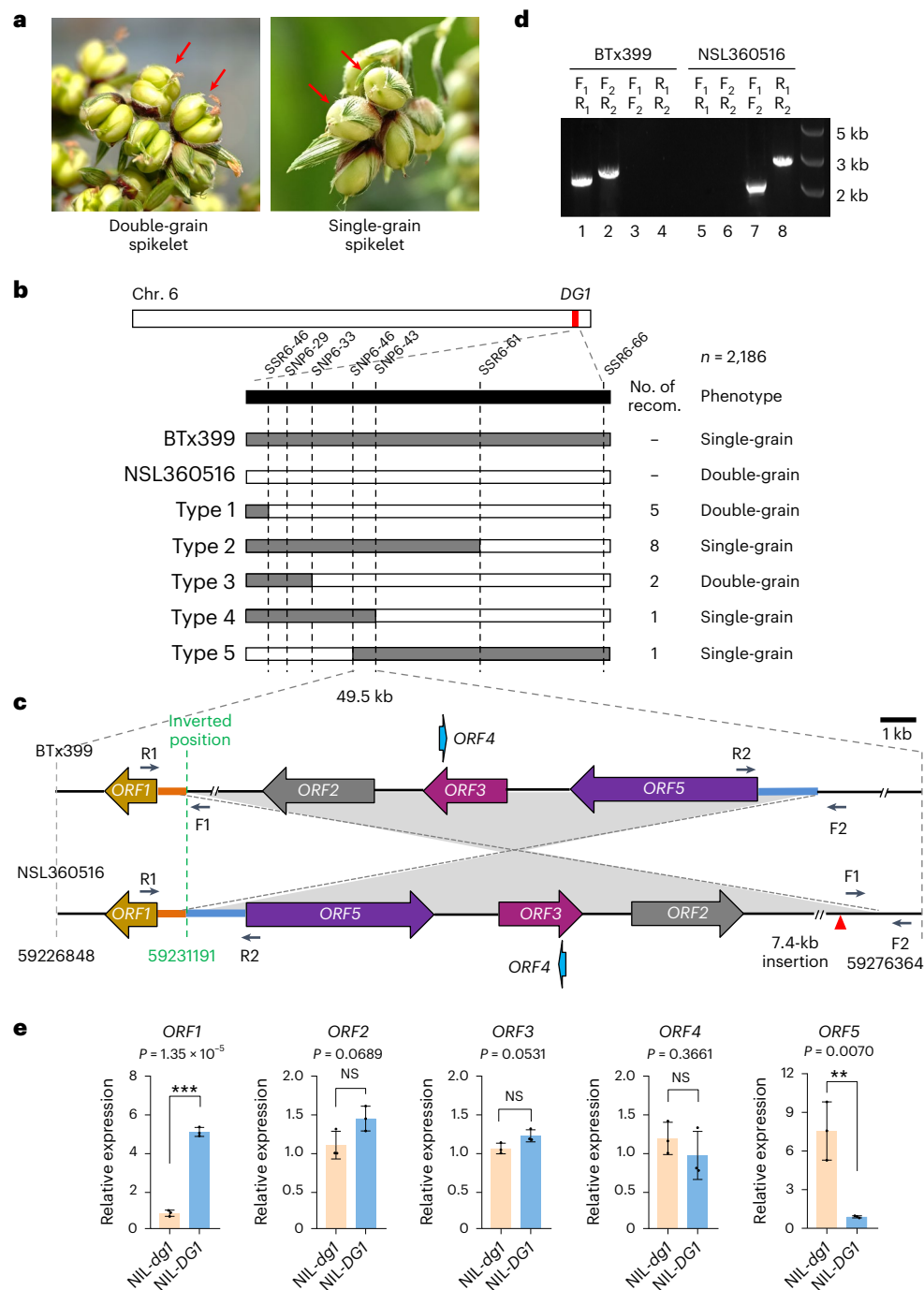


Fig. 1 | Identification and map-based cloning of the *DG1* locus in controlling the sorghum double-grain trait. a, The morphology of double-grain spikelets and single-grain spikelets in sorghum. **b**, Initial mapping and fine mapping of the *DG1* locus. Analysis of 875 individuals in the F₂ generation from NSL360516 and three single-grain lines revealed the mapping of the *DG1* locus between markers SSR6-46 and SSR6-66 at the end of chromosome 6. Using 2,186 plants from F₃ and F₄ populations from the cross between NSL360516 and BTx399, we ultimately localized the *DG1* locus to a 49.5-kb region between markers SNP6-46 and SNP6-43 through the analysis of 17 recombinants. **c**, Sequence alignment of five annotated genes within the *DG1* region. A 35.7-kb chromosomal paracentric inversion

occurred in NSL360516. The primers F₁, R₁, F₂ and R₂ were designed for the confirmation of the inversion. A 7,401-bp insertion was identified in the inverted segment of NSL360516, highlighted by a red triangle. **d**, PCR products were amplified using primers F₁/F₂ and R₁/R₂ in NSL360516. Additional PCR products were amplified using primers F₁/R₁ and F₂/R₂. DNA bands 1, 2, 7 and 8 were also purified and sequenced to verify the inversion. Four independent experiments were carried out. **e**, The relative expression levels of five ORFs in young panicles of NIL-*dg1* and NIL-*DG1* by qPCR analysis. The data are shown as mean ± s.d.; n = 3 biological replicates. The P values were calculated via two-tailed Student's *t*-tests. ****P* < 0.001; ***P* < 0.01; NS, no significant difference.

than 2% background difference between the NILs was observed using 277 SSR markers across the entire genome (Extended Data Fig. 2b). To identify the candidate for the *DG1* locus, we conducted gene expression analysis among the five genes in NILs through quantitative real-time PCR (qRT-PCR). The results indicated no significant difference in the

ORF2, *ORF3* and *ORF4* genes between NIL-*dg1* and NIL-*DG1*. However, *ORF1* showed significant upregulation and *ORF5* exhibited apparent downregulation in NIL-*DG1* compared with NIL-*dg1* (Fig. 1e).

We subsequently focused on *ORF1* and *ORF5* due to the observed variations at both the DNA and mRNA levels. Transgenic plants were

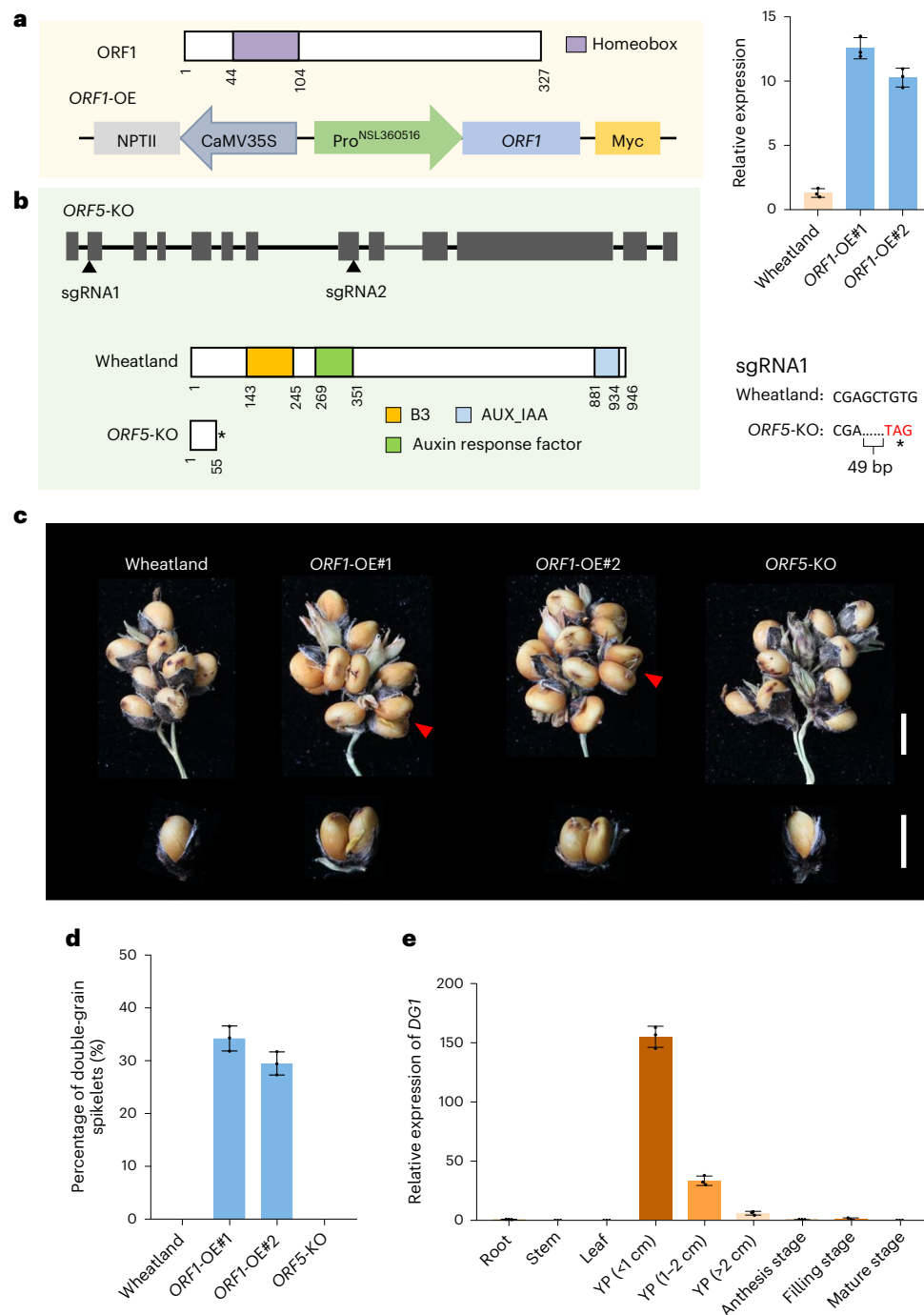


Fig. 2 | *ORF1* is the candidate gene for *DGI*. **a**, The peptide structure of *ORF1*. The *ORF1*-OE vector contains the coding sequence of *ORF1* driven by the native promoter of *NSL360516*, with a Myc tag located at the C terminus. The construct was transformed into the sorghum recipient line (Wheatland). Gene expression of *DGI* in two independent lines, *ORF1*-OE#1 and *ORF1*-OE#2, in the T_0 generation was detected. **b**, The two sgRNA targets and peptide sequence alignment for *ORF5* and the *ORF5*-KO mutant. A premature stop codon obtained in *ORF5*-KO is shown in red, along with an asterisk. **c**, Spikelet morphology of Wheatland,

ORF1-OE lines (*ORF1*-OE#1 and *ORF1*-OE#2) and *ORF5*-KO. Scale bars, 0.5 cm. The red arrowheads indicate double-grain spikelet. **d**, Statistical data on double-grain spikelets in Wheatland, *ORF1*-OE and *ORF5*-KO. The data are shown as mean $\pm$ s.d.; $n = 3$ biological replicates. **e**, Relative expression of *DGI* in various tissues, including root, stem, leaf, young panicles, panicle after flowering, seeds at the grain filling stage and mature stage. The data are shown as mean $\pm$ s.d.; $n = 3$ biological replicates.

constructed using overexpression or knockout approaches in Wheatland, a recipient sorghum line with a single-grain phenotype, facilitated by *Agrobacterium*-mediated genetic transformation. Given the higher expression of *ORF1* in NIL-*DGI*, we generated an *ORF1*-overexpressing (*ORF1*-OE) vector driven by a native ~2.5-kb promoter from *NSL360516*. Two independent positive *ORF1*-OE plants were obtained in the T_0

generation (Fig. 2a). At the mature stage, both the *ORF1*-OE#1 and *ORF1*-OE#2 lines displayed the double-grain trait with percentages of 34.2% and 29.5%, respectively (Fig. 2c,d). We previously attempted *ORF1* knockout in double-grain lines, but the transformation failed due to the challenging genetic background. Due to the downregulation of *ORF5* in NIL-*DGI*, we created an *ORF5*-knockout (*ORF5*-KO) mutant through

CRISPR–Cas9 technology. A homozygous *ORF5*-KO line, free of the Cas9 carrier, was obtained in the T₂ generation. *ORF5*-KO exhibited a ~2-kb chromosomal paracentric inversion between the two designed single guide RNA (sgRNA) targets, resulting in the acquisition of a premature stop codon at +56 amino acids (Fig. 2b and Extended Data Fig. 3). Each spikelet in *ORF5*-KO displayed a single-grain trait, the same as the Wheatland recipient line (Fig. 2c,d). We thus concluded that *ORF1* (Sobic.006G254900) is the candidate gene for *DG1* and plays a positive regulatory role in the double-grain trait.

DG1 encodes a putative homeobox-domain-containing transcription factor (Fig. 2a and Extended Data Fig. 4a). Phylogenetic analysis revealed that *DG1* is an orthologue of *AtWUS* in *Arabidopsis*, *ZmWUS1/2* in maize and *OsWUS* in rice (Extended Data Fig. 4b). Spatiotemporal expression pattern analysis showed that *DG1* was highly expressed in the early young panicles (0–1-cm young panicle stage) and decreased progressively in the following periods (Fig. 2e). As there were no variations in the encoding region between *dg1* and *DG1*, our attention subsequently shifted to the inversion in the promoter, which might alter promoter activity.

***DG1* enhances expression by reducing histone methylation**

Structural variations in gene promoters exemplify key drivers in plant evolution, modulating gene expression to produce advantageous phenotypes that enable adaptation to shifting environmental conditions without modifying the coding sequence of genes. Moreover, large-scale structural changes can impact chromatin accessibility in promoter regions, potentially altering gene expression through mechanisms such as reduced repressive methylation modifications^{27,29}. H3K27me₃, a repressive epigenetic modification commonly found near transcriptional start sites, enforces transcriptional repression³⁰. To explore this relationship, we examined the potential correlation between *dg1* gene expression and methylation modifications. Data from a public repository (<http://epigenome.genetics.uga.edu/PlantEpigenome/>)³¹ indicated that two regions exhibit high levels of H3K27me₃ (Fig. 3a), spanning from 59230360 to 59233365 bp (0 to ~3 kb upstream of the *dg1* translation start codon) and from 59240020 to 59247220 bp (~9.7 to ~16.9 kb upstream of the *dg1* translation start codon). In addition, H3K9me₂, a heterochromatin marker enriched at transposable elements (such as *Gypsy* LTRs in promoter regions), often exerts repressive effects on the expression of downstream genes^{32,33}. We also detected several transposable elements, notably a ~6.2-kb RLG_scAngela_2_1-LTR retrotransposon belonging to the typical *Gypsy* superfamily, flanking the two highly methylated regions in the *dg1* promoter, which may contribute to the observed methylation. We reasoned that the 35.7-kb paracentric inversion in *DG1* may alter the homeostasis of epigenetic modifications, thereby altering the transcriptional activity of the promoter.

To test this hypothesis, we determined the accumulation of H3K9me₂ and H3K27me₃ in the young panicles of NILs at three developmental stages via chromatin immunoprecipitation with quantitative PCR (ChIP–qPCR) assays. Three different regions were designated: R1, R2 and R3, located at positions ~150 bp, ~300 bp and ~1.7 kb, respectively. R4 (~21.7 kb) and R5 (~38 kb), positioned in the middle and far end of the inversion fragment, served as the negative controls. The results indicated that, compared with NIL-*dg1*, NIL-*DG1* exhibited a great decrease in the H3K27me₃ level across all young panicle stages (Fig. 3b). By contrast, H3K27me₃ levels at R4 and R5 were only slightly altered or not changed. Additionally, the H3K9me₂ level in the 0–1-cm young panicles of NILs was assessed, revealing a significant reduction in H3K9me₂ levels in *DG1* compared with *dg1* (Fig. 3c). These findings suggest that the natural inversion in *DG1* leads to a significant downregulation of the enrichment of repressive epigenetic modifications, including both H3K27me₃ and H3K9me₂, near the transcription start site of *DG1*, thereby derepressing the transcriptional inhibition of *DG1*.

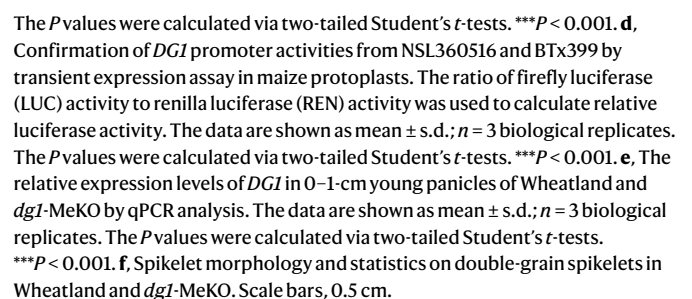
To examine the impact of alterations in the first highly methylated region on gene expression, we cloned approximately 4.0-kb promoter fragments of *DG1/dg1* from NSL360516 and BTx399 into the pGreenII-0800–LUC vector for a transient expression assay in maize protoplasts. The results showed that Pro^{NSL360516}::LUC indeed had significantly increased luciferase activity compared with Pro^{BTx399}::LUC (Fig. 3d). Even though SNPs or insertions and deletions within the promoter sequence cannot be excluded as potential elements that have an impact on gene expression, the low-methylation region in the new promoter is probably a key contributor to the upregulation of *DG1* expression. To further investigate the role of H3K27me₃ modification in regulating the sorghum double-grain trait, we generated a sorghum mutant line using CRISPR–Cas9 technology. In the T₂ generation, the homozygous mutant, *dg1*-MeKO, harboured a deletion encompassing approximately 23.3 kb from position 59241800 bp to 59265140 bp, resulting in the loss of the second highly methylated region at its promoter. We assessed *DG1* expression at the young panicle stage and observed a significant increase in transcriptional levels in *dg1*-MeKO compared with the control (Fig. 3e). At maturity, *dg1*-MeKO exhibited a distinct double-grain phenotype (with a double-grain spikelet percentage of 22.9%) from that of the wild type, Wheatland (Fig. 3f).

***DG1* restores the fertility of the lower floret and floret meristem activity**

In the typical developmental progress of a dual-floret-type spikelet, such as the SS in sorghum, the outer and inner glume primordia are initially formed along the base of the spikelet meristem. Subsequently, the first lemma primordium and the first floret primordium at the lower position of the spikelet begin differentiation almost simultaneously. This lower floret typically undergoes arrested development and quickly disappears at an early stage¹. At this moment, the second lemma primordium is generated, followed by the emergence of a second floret primordium at the upper position. This upper floret proceeds to develop into a mature grain. So far, the reason for the initial lower floret in a spikelet undergoing abortion remains unclear.

The developmental mechanism underlying the formation of the double-grain morphology may shed light on this question. We observed various stages of young spikelets in NILs through scanning electron microscopy. In both NILs, the floret meristem and a pair of glume primordia formed at stage 1, followed by the initiation of the lemma primordium and the differentiation of the upper and lower floret meristem at stage 2. However, the lower floret primordium rapidly vanished in NIL-*dg1*, whereas it remained clearly visible adjacent to the first lemma in NIL-*DG1* at stage 2. In both NILs, the upper floret primordium developed into three stamens and a pistil primordium from stage 3 onward. However, only in NIL-*DG1* did the lower floret primordium continue differentiation during stages 3 and 4, lagging behind the second upper floret but ultimately developing into a double-grain spikelet together with the upper floret (Fig. 4a).

Paraffin section analysis at stages 2 and 3 confirmed the presence of a fertile lower floret in NIL-*DG1*. In NIL-*dg1*, only the second upper floret primordium maintained meristem activity. However, in NIL-*DG1*, in addition to the normal upper floret, the lower floret primordium near the outer glume and first lemma was distinctly observed (Fig. 4b). RNA in situ hybridization in NIL-*DG1* spikelets revealed that *DG1* was predominantly expressed in the floret meristem during stage 1. At stage 2, *DG1* exhibited high expression at the apical meristem region of both the upper and lower floret primordia (Fig. 4c). These findings suggest that *DG1* may play an important role in floret meristem initiation and differentiation. As the NILs reached the flowering stage, NIL-*dg1* spikelets contained a normal pistil and three stamens. In contrast, NIL-*DG1* spikelets exhibited the induction of two pistils and more than three (four to six) stamens, while the upper floret continued to produce the normal number of floral organ primordia. (Fig. 4d and Extended Data Fig. 5a,b). The additional pistil and stamens originate from the



843 and 643 differentially expressed genes (DEGs) were identified at the YP1 and YP5 stages, respectively (Fig. 5a). Our primary focus was on the DEGs at YP1 due to their notably high expression level and spatial expression pattern at S2 and S3 (Figs. 2e and 4c). Gene Ontology (GO) enrichment analysis revealed that a notable proportion of DEGs were enriched in the pathways related to sequence-specific DNA-binding transcription factors and proteins functioning in hormone regulatory pathways (Fig. 5b).

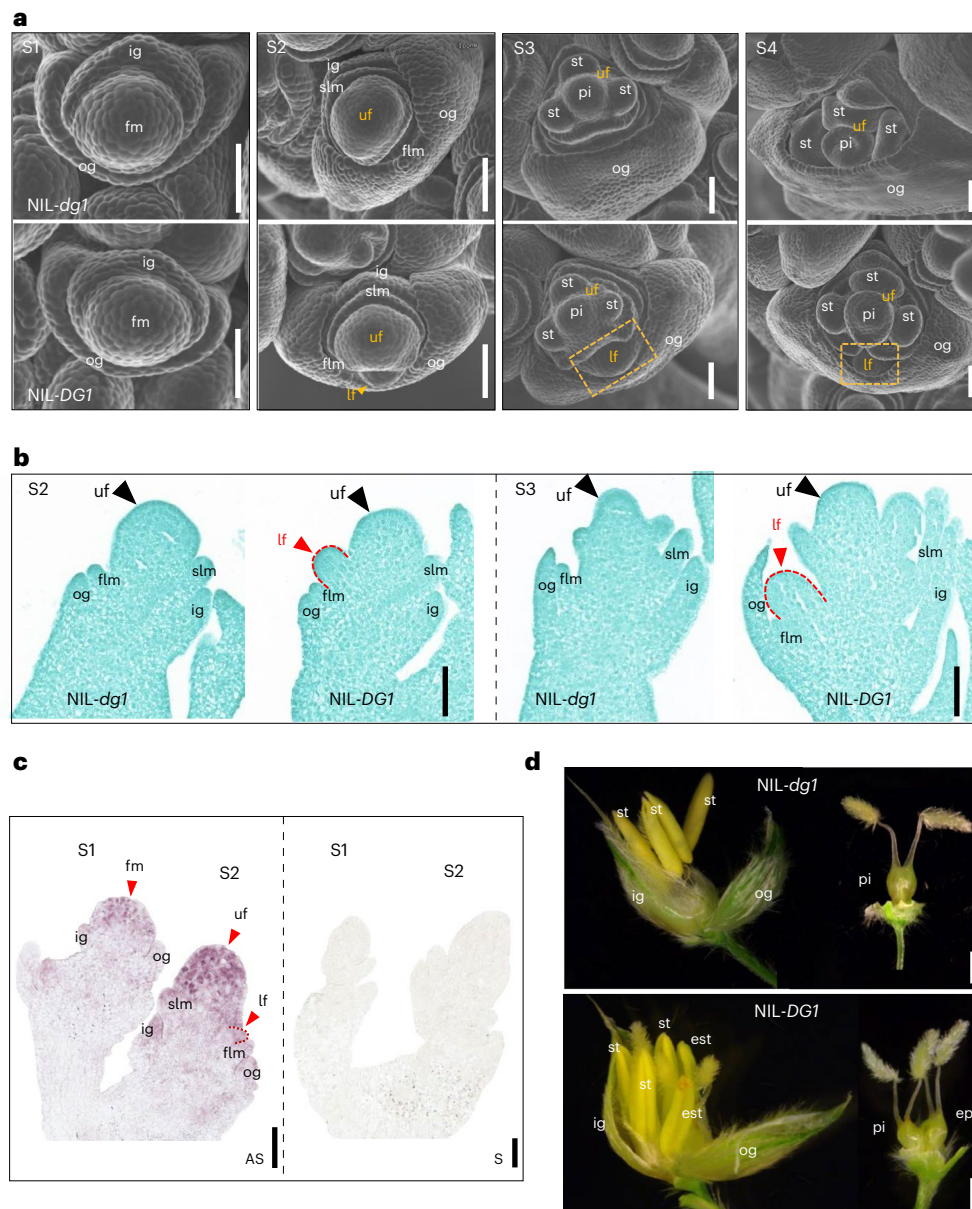


Fig. 4 | *DGI* exhibits fertility of the lower floret to form double grains.

a, Scanning electron micrographs of spikelets at different stages in *NIL-dg1* and *NIL-DG1*. S1, S2, S3 and S4 indicate the four stages of spikelet developmental processes within <2-cm young panicles. Scale bars, 50 μ m. For each line, 500 spikelets from 20 individual plants were observed. The orange rectangular box represents the lower floret whose fertility has been restored. **b**, Paraffin section analysis of floret meristem of *NIL-dg1* and *NIL-DG1* at the S2 and S3 stages. The first lower floret is highlighted in red. Scale bars, 50 μ m. **c**, RNA in situ hybridization of *DGI*. Longitudinal sections of spikelets at the S1 and S2

stages were analysed. A negative control was performed using a sense probe. AS, antisense probe; S, sense probe. Scale bars, 50 μ m. **d**, Phenotypes of spikelets in *NIL-dg1* and *NIL-DG1* at the flowering stage. For each line, 200 spikelets were examined. In addition to the three stamens and one pistil of *NIL-dg1*, *NIL-DG1* spikelets formed extra stamens and an additional pistil. Scale bars, 1 mm. og, outer glume; ig, inner glume; fm, first lemma; lf, lower floret; uf, upper floret; flm, first lemma; slm, second lemma; st, stamen; pi, pistil; est, extra stamen; epi, extra pistil.

The transcription factors mainly include the Basic-leucine zipper (bZIP), APETALA2 (AP2) and MADS families, which are involved in floral organ morphogenesis. The expression of Sobc.006G210300, the homologue of *OsAP2-39*, was also upregulated in *NIL-DG1* (Extended Data Fig. 6b). Meanwhile, two MADS-box genes, Sobc.010G261800 (the homologue of *OsMADS16*) and Sobc.001G314400 (the homologue of *OsMADS56*), were both downregulated in *NIL-DG1*. *DGI* also elevated the expression of Sobc.007G082200 (Extended Data Fig. 6c), the homologue of rice *OsABCG26*, which functions in anther development and pollen activity. The *Osabcg26* mutant develops small and pale anthers, as well as defects in the anther cuticle³⁷. These results indicate that *DGI* produces more stamen architecture through regulating *SbbZIP34*, *SbAP2-39*, *SbMADS16*, *SbMADS56* and *SbABCG26* (Fig. 5c,d).

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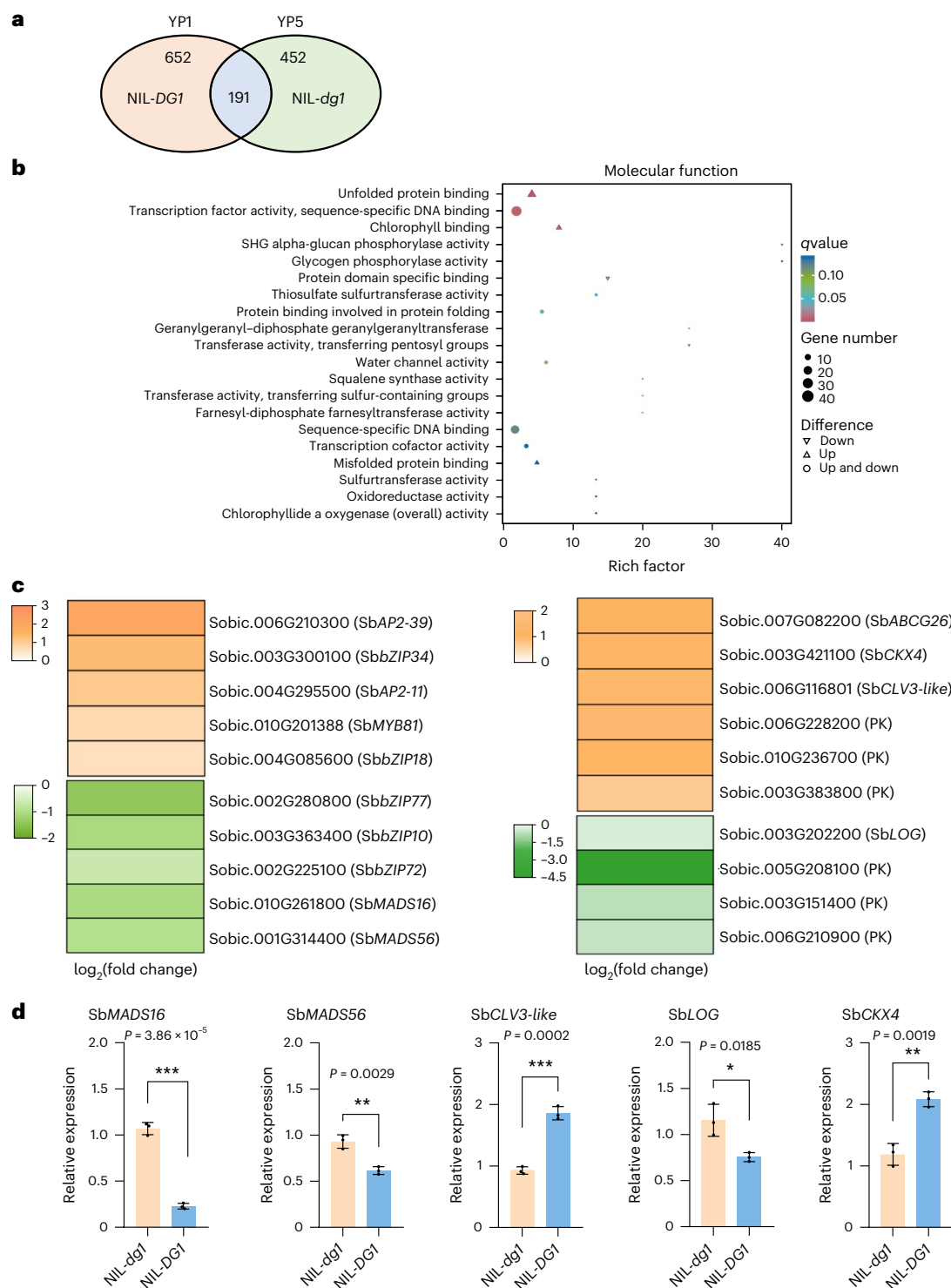


Fig. 5 | The potential downstream pathways of *DGI*. **a**, Venn diagram illustrating the number of DEGs between *NIL-dg1* and *NIL-DG1* at two developmental stages, YP1 and YP5. **b**, GO term enrichment analysis of *DGI*-regulated pathways at the YP1 stage. Significantly enriched GO terms were determined on the basis of an adjusted $P < 0.05$. **c**, Expression patterns of DNA-binding transcription factors (left) and critical genes involved in floral meristem activity (right) during the YP1 stage. Upregulated genes in *NIL-DG1* are represented in orange,

while downregulated genes are indicated in green. PK, protein kinase. **d**, Gene expression detection of *SbMADS16*, *SbMADS56*, *SbCLV3-like*, *SbLOG* and *SbCKX4* (the homologues of rice *OsMADS16*, *OsMADS56*, *AtCLV3*, *OsLOG* and *OsCKX4*, respectively) in *NIL-dg1* and *NIL-DG1* during the YP1 stage using qPCR assays. Three biological repeats were performed. The P values were calculated via two-tailed Student's t -tests. *** $P < 0.001$; ** $P < 0.01$; * $P < 0.05$.

Previous studies have shown that *WUS* can directly activate the expression of *CLV3* to control IM homeostasis in plants^{18,38,39}. *Sobic.006G116801*, a gene encoding a CLV3-like CLE protein, was significantly upregulated by *DGI*, suggesting a conserved negative-feedback

regulation of the *WUS*–*CLV3* pathway in sorghum double-grain formation. Cytokinin also plays a critical role in the promotion of floral meristem maintenance and termination⁴⁰. *LONELY GUY (LOG)* encodes a cytokinin-activating enzyme and acts as the final step of cytokinin

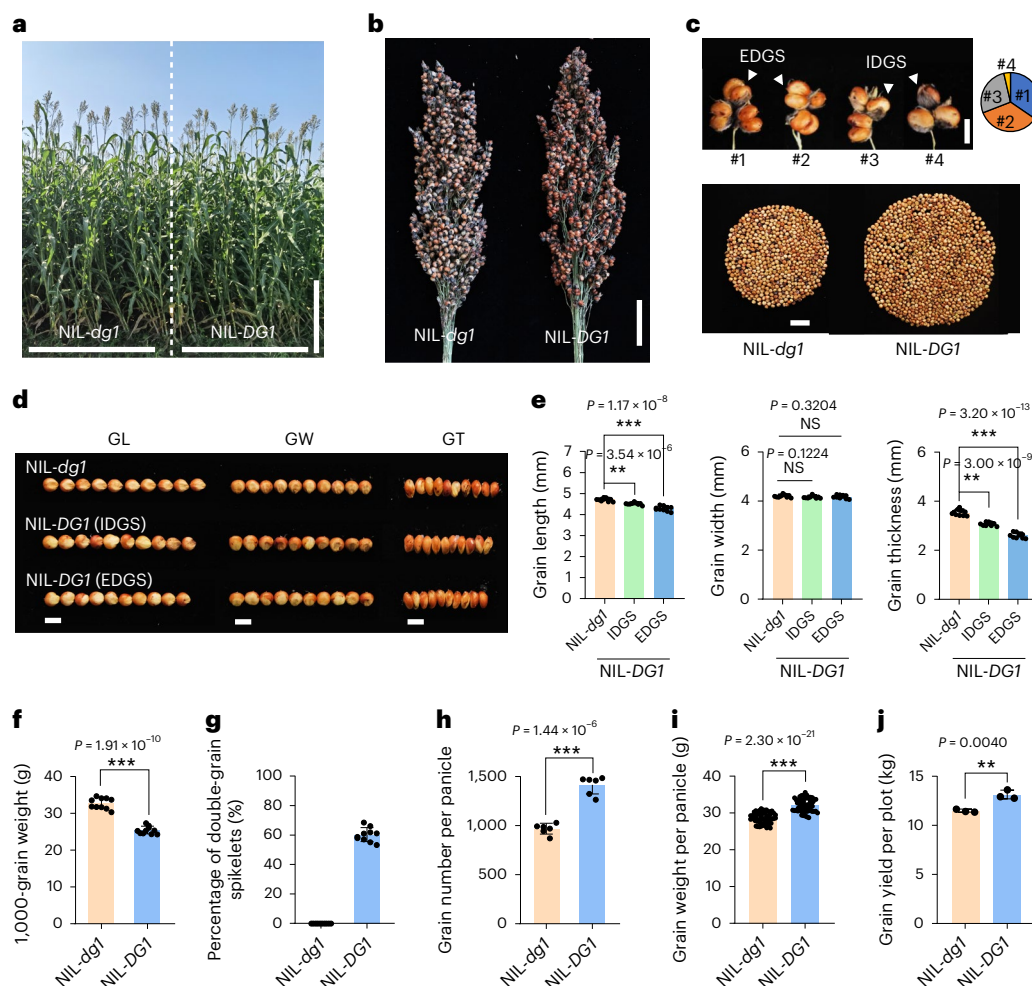


Fig. 6 | *DG1* potentially enhances grain number and grain yield. a, Field plant architecture of NIL-*dg1* and NIL-*DG1*. Field yield tests were conducted in Beijing in 2024. Scale bars, 1 m. **b**, Representative mature panicle morphology of NIL-*dg1* and NIL-*DG1*. Scale bar, 5 cm. **c**, Top, the ratio of the four types of tertiary branches containing three spikelets in NIL-*DG1*. Bottom, the morphology of grain number per panicle in NILs after threshing. Upper and lower scale bars, 5 mm and 2 cm, respectively. **d**, Comparison of grain length (GL), grain width (GW) and grain thickness (GT) among single-grain spikelets of NIL-*dg1* and IDGSs and EDGSs of NIL-*DG1*. $n = 10$. Scale bars, 5 mm. **e**, Statistical analysis of ten-grain

length, grain width and grain thickness among single-grain spikelets of NIL-*dg1* as well as IDGSs and EDGSs of NIL-*DG1*. **f–j**, Statistical analysis of 1,000-grain weight ($n = 10$), ratio of double-grain spikelets (EDGS, $n = 10$), grain number per panicle ($n = 6$), grain weight per panicle ($n = 60$) and grain yield per plot ($n = 3$; plot size was 10 m long and 4 m wide, with row spacing set at 50 cm and plant spacing at 20 cm; also see Methods) in NIL-*dg1* and NIL-*DG1*. The data are shown as mean $\pm$ s.d. The P values were determined via one-way analysis of variance with Tukey's multiple comparisons test and two-tailed Student's t -tests. *** $P < 0.001$; ** $P < 0.01$; * $P < 0.05$.

biosynthesis to control meristem activity. In rice and *Arabidopsis*, loss of function of *LOG* genes leads to the premature termination of the shoot apical meristem^{41–43}. In maize, higher *ZmWUS1* expression in the *Bif3* mutant caused severe damage to the cytokinin biosynthesis pathway, leading to architectural rearrangements of IMs and resulting in ear meristem overproliferation²³. Cytokinin oxidase/dehydrogenase (CKX) is a group of specific enzymes that reduce cytokinin content through oxidative cleavage⁴². Overexpression of cytokinin oxidase, leading to reduced cytokinin levels, results in meristem defects⁴⁴. We observed that *Sobic.003G202200* (*SbLOG*, an orthologue of rice *OsLOG*) was significantly downregulated, while *Sobic.003G421100* (*SbCKX4*, an orthologue of rice *OsCKX4/REN1*) was significantly upregulated in NIL-*DG1* compared with NIL-*dg1* (Fig. 5c,d), possibly indicating a negative feedback loop of cytokinin synthesis by *DG1* during the double-grain developmental process. These data suggest that floral meristem maintenance by the negative feedback crosstalk between WUS–CLV3 and cytokinin pathways is required for sorghum double-grain generation.

Potential utilization of *DG1* for sorghum yield improvement

Grain number per panicle and 1,000-grain weight are crucial factors influencing crop grain yield. In the case of sorghum, with the formation of two mature grains in the spikelet, the introduction of *DG1* was hypothesized to enhance grain yield. Multi-year (2023 and 2024) and multi-location field-yield trials (Beijing and Shenzhen cities) of the NIL lines were conducted. We observed no clear differences among the plant-architecture-related traits, such as plant height, tillering and flowering time (Fig. 6a). However, at the mature stage, NIL-*DG1* panicles showed a notably higher grain number than NIL-*dg1*, largely due to the increased proportion of double-grain spikelets (EDGSs) positioned on the tertiary branches (Fig. 6b,c and Extended Data Figs. 7a and 8a). Previous studies found that multiple-grain spikelets may decrease the grain size and grain weight in wheat and sometimes result in abnormal grains in rice mutants^{45,46}. We therefore measured the grain size and weight-related traits in all spikelets of NILs. Compared with NIL-*dg1*, both IDGSs and EDGSs in NIL-*DG1* showed reductions in grain length by 4.1% to 9.5% and 8.9% to 13.4%, respectively, and decreases in grain

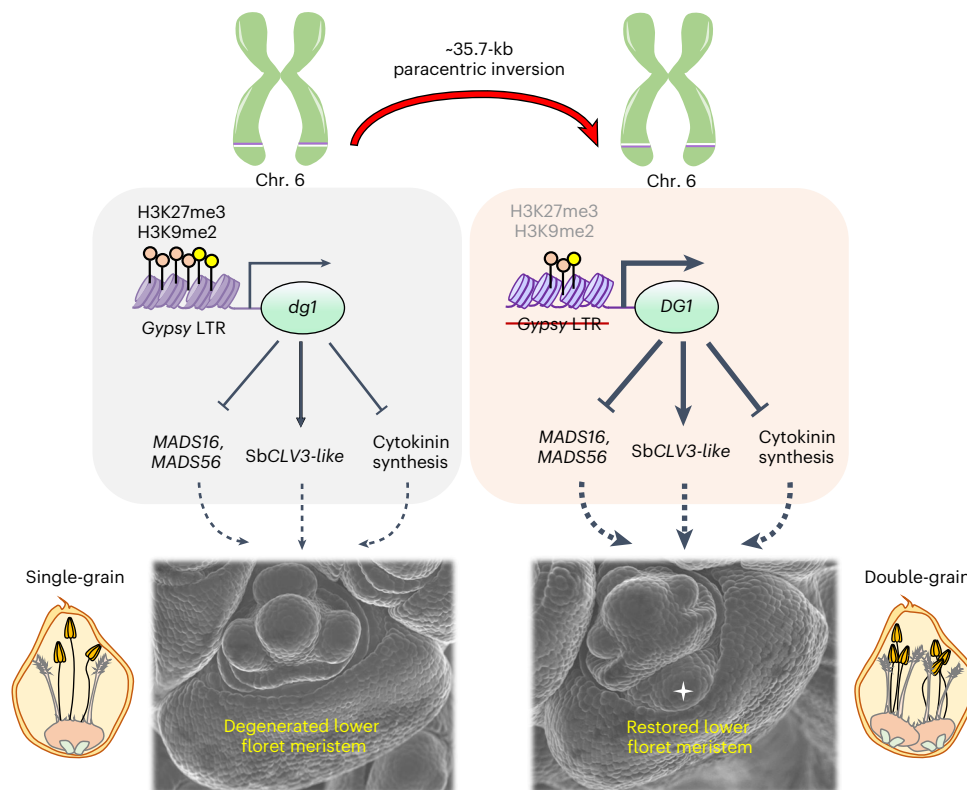


Fig. 7 | A proposed working model for the regulation of the double-grain trait by *DGI* in sorghum. A chromosomal paracentric inversion, spanning ~35.7 kb, took place in the promoter region of *DGI*, situated at the end of chromosome 6. This modification led to an increase in *DGI* expression by eliminating a ~6.2-kb *Gypsy* LTR and reducing the levels of H3K27me3 and H3K9me2, facilitating the transformation of the degenerated lower floret into a fertile one. This promoted

the additional development of pistils and stamens, ultimately leading to the formation of the double-grain trait. During this developmental process, genes such as *MADS16* and *MADS56*, *SbCLV3-like*, and cytokinin synthesis pathway genes are significantly regulated by *DGI*. These genes may play a crucial role in restoring the activity of the lower floret meristem.

thickness by 12% to 13.1% and 18% to 26.2%, respectively (Fig. 6d and Extended Data Fig. 7b). However, there were no significant differences observed in grain width between the two lines (Fig. 6e and Extended Data Fig. 7c). The 1,000-grain weight of NIL-*DGI* was reduced by 20.8% to 23% compared with NIL-*dg1* across both field-yield trials (Fig. 6f and Extended Data Figs. 7d and 8b).

Despite the adverse impacts on grain weight, we observed that 56.7% to 57.3% of spikelets in NIL-*DGI* exhibited enhanced double-grain spikelets (EDGs), leading to an approximate increase of 40.7% to 46.1% in grain number per panicle compared with NIL-*dg1* (Fig. 6g,h and Extended Data Figs. 7e,f and 8c). This resulted in an 8.6% to 12.4% increase in grain weight per panicle in NIL-*DGI* compared with NIL-*dg1* (Fig. 6i and Extended Data Figs. 7g and 8d). These findings indicate that the natural *DGI* allele could serve as a valuable resource for improving grain number and yield in sorghum. In field trials involving a large number of NILs, the overall grain yield per plot of NIL-*DGI* was found to be 10.1%, 14.3% and 12.7% higher than that of NIL-*dg1* in the 2023 Beijing trial, the 2024 Beijing trial and the 2024 Shenzhen trial, respectively (Fig. 6j and Extended Data Figs. 7h and 8e), underscoring the potential of *DGI* for enhancing sorghum yield. Given the substantial positive impact of *DGI* on grain number, future molecular design efforts to integrate additional high-grain-weight genes could greatly boost the grain yield of sorghum.

Discussion

The development of spikelet and floret fertility plays a crucial role in determining grain number and grain yield in grasses. A spikelet, the fundamental unit of inflorescence in grasses, comprises one or more florets depending on species-specific characteristics⁴⁷. For instance,

barley is characterized by a single-floret spikelet, while wheat spikelets typically encompass three to nine florets. The ancestors of rice potentially feature a three-floret spikelet^{1,48}. Sorghum grain-bearing spikelets give rise to a dual-floret type. Typically, the lower floret undergoes abortion, while the upper floret develops normally and produces a single grain. In cereals, researchers have observed the emergence of multiple-grain traits in certain domesticated varieties and mutant materials^{2,22,46,49–52}. In this study, we have unveiled a major locus, *DGI*, that governs the dual-floret spikelet architecture in sorghum. A unique chromosomal paracentric inversion in the promoter region of *DGI* was identified, leading to enhanced gene expression and the manifestation of the double-grain trait. The natural paracentric inversion potentially reduces both H3K27me3 and H3K9me2 levels by removing the *Gypsy* LTR fragment. The alteration in the homeostasis of epigenetic modifications probably enhances *DGI* expression and promotes lower floret meristem activity in sorghum. Similarly, natural chromosomal structural variations, such as inversions, deletions and insertions upstream of the *BRD3* promoter, can activate its expression and inhibit the tissue-specific brassinosteroid pathway, leading to over a 40% increase in grain number in clustered spikelet rice^{53,54}. *DGI* was found to be highly expressed in early young panicles, suggesting its involvement in the regulation of floret meristem initiation and differentiation. The *DGI* allele, characterized by reduced repressive histone lysine methylation and increased expression, has been demonstrated to restore fertility to typically sterile lower florets, thereby leading to the double-grain trait observed in sorghum spikelets (Fig. 7).

The CLV–WUS negative feedback loop is a key regulatory mechanism functioning in plant shoot apical meristem and IM maintenance^{15,55}. *DGI*, encoding a homeobox domain transcription

factor, is an orthologue of maize *ZmWUS1* and *Arabidopsis AtWUS* and is involved in floret meristem activity in sorghum. Previous studies have shown that *AtWUS* plays a key role in regulating IM activity, while *ZmWUS1* is crucial for maintaining inflorescence and stem meristem size in maize^{17,23}. Loss of function in *CLV1*, *CLV2* or *CLV3* along with enhanced *WUS* expression leads to IM overproliferation, resulting in architectural rearrangements of spikelet and floret meristems as well as the formation of extra floral organs^{18,56,57}. *OsWUS/TAB1* plays an important role in tiller bud formation but is not involved in promoting stem cell proliferation in the shoot apical meristem and IM^{58,59}. Rice *tg1*, a novel gain-of-function allele of rice *fon2/fon4*, may act through a similar pathway in regulating double-grain spikelets and exhibits a 24.1% increase in grain yield². The *tg1* floral meristem shows an increased number of carpels and can develop two or even three grains per spikelet, similar to *DG1*. In maize, severe mutations in the orthologues of *Arabidopsis CLV*-like genes, including *THICK TASSEL DWARF 1*, *FASCIATED EAR2 (FEA2)*, *EMBRYO SURROUNDING REGION-RELATED 7 (ZmCLE7)* and *FON2-LIKE CLE PROTEIN 1 (ZmFCP1)*, have been associated with enlarged inflorescences and fasciated ears, resulting in reduced yield^{60–63}. However, weak alleles of *fea2* and *fea3*, with moderate upregulation of the *WUS* gene, led to an increase in kernel rows and maintained normal inflorescence, thereby enhancing yield^{63–65}. In this study, we observed that *DG1* does not affect IM size (these data were not shown) but leads to notable differences in floret meristem initiation and differentiation. The naturally elevated expression of sorghum *DG1* positively enhances lower floret meristem activity and restores the fertility of previously sterile florets, highlighting the role of *WUS* in regulating floret meristem rearrangement and floral organ development in sorghum. We did not observe significant differences in inflorescence architecture caused by *DG1*. However, changes in floret primordia activity alone by *DG1* significantly increased the number of grains per sorghum panicle, ultimately leading to higher yields in sorghum.

In fact, approximately 40% of spikelets in NIL-*DG1* exhibit the production of IDGS structures, characterized by an additional underdeveloped grain at the mature stage. Upon observing the grain morphology of IDGSs, it is evident that the additional floret cannot develop into a mature grain, and the other is slightly smaller than the single-grain seed. In EDGSs, both mature grains are normal but obviously smaller than the single-grain line. We speculate that this is primarily caused by an inadequate energy supply during panicle growth, impacting grain size and yield. The conflicting effects of an increase in grain number and a decrease in grain size are evident. Yield tests demonstrate that the excellent allele, *DG1*, ultimately increases the final grain yield by 10.1% to 12.4% when compared with *dg1*. Similarly, the rice *tg1* significantly improves grain yield to a larger extent². These findings suggest that *DG1* holds potential for high-yield breeding by substantially increasing grain number. Further integration of other major genes, such as *qGW1* and *qTGW1a*^{66–68}, the major loci for grain width and grain length in sorghum, may enhance grain size and mitigate grain weight loss.

The number of florets within a spikelet is a key factor influencing grain number in crops. Most cereals, such as sorghum, millet, maize and coix, possess a two-floret spikelet, whereas wheat and wild rice exhibit a multiple-floret spikelet. In the dual-floret spikelet, the upper floret is consistently fertile, while the lower floret is sterile, often degenerating into a lemma- or glume-like structure¹. In rice, the ‘three-floret spikelet’ hypothesis suggests that the ancestral spikelet originally had two lateral florets along with a central fertile floret. However, the two lateral florets in cultivated varieties have degenerated into sterile lemmas, resulting in a single grain per spikelet. Researchers have been attempting to clone the genes responsible for sterile lemma identity and lateral floret meristem activity—such as *LF1*, *MFO1*, *G1*, *LRG1/NSG1*, *DF1/EG1*, *ASPI*, *OsMADS34*, *SAD1* and *OsMADS1*—to recreate the three-floret spikelet and increase grain number^{48,69–75}. Here our discovery of *DG1* function in sorghum demonstrates the potential for lateral florets in double-grain spikelets to fully develop into mature grains, providing

new insights into the three-floret spikelet in rice. The degeneration mechanism of lateral florets in rice remains unique, probably requiring the coordinated regulation of multiple genes to achieve the production of multiple fertile florets.

The market value of sorghum grain extends beyond its use in food; it is also a crucial raw material for brewing liquor (Baijiu) in China⁷⁶. Many renowned Chinese liquors, such as Moutaijiu, Wuliangye, Fenjiu and Langjiu, are brewed using sorghum grain, making it a favoured choice among consumers⁷⁷. Sorghum liquor production employs two fermentation methods: whole-grain fermentation and crushed-grain fermentation. In liquors such as Moutaijiu that use the whole-grain fermentation method, the size and 1,000-grain weight of sorghum grains can impact the fermentation process. A small, flattened grain shape is expected to enhance fermentation efficiency and reduce susceptibility to damage during multiple rounds of cooking. The double-grain trait controlled by *DG1*, with its higher grain number and smaller grain size, also holds potential for enhancing Chinese liquor sorghum breeding.

Methods

Materials

A diverse panel of 571 sorghum accessions, sourced from the United States Department of Agriculture–Agricultural Research Service, was used to investigate the double-grain trait. These accessions were cultivated in both Beijing (39.5° N, 116.4° E) and Sanya (18.3° N, 108.8° E). Two lines identified with the double-grain trait, NSL360516 and F.R. Miller, were crossed with single-grain lines (SC855, KS115 and BTx399) to produce the F₁ generation plants. Three F₂ populations, comprising a total of 875 individuals, were generated through self-pollination of F₁ plants resulting from crosses between NSL360516 and the single-grain accessions. The F₂ populations derived from NSL360516 and three single-grain accessions were employed for the initial mapping of *DG1*. Two expanded F₃ and three F₄ segregated populations were developed from NSL360516 and BTx399 for fine mapping *DG1*, which was eventually pinpointed to a 49.5-kb region flanking the markers *SNP6-46* and *SSR6-43*. Through marker-assisted selection and self-crossing, a pair of NILs of *DG1* was established in the F₇ generation. All parental lines and populations were planted in soils with a row interval of 50 cm and a plant interval of 20 cm.

Field-yield statistical experiments were conducted at Changping Farm, Beijing, China (116.37° E, 40.10° N) during the summers of 2023 (Yield Test I) and 2024 (Yield Test II), as well as in Shenzhen city, China (114.00° E, 22.80° N) in the summer of 2024 (Yield Test III). Three plots for the NILs were randomly designed. For Yield Tests I and II, the planting plot size (approximately 6,600 plants per mu) for the NIL lines was 10 m long and 4 m wide, while for Yield Test III, the plot size was 4 m long and 3 m wide. Row spacing was 50 cm, and plant spacing was 20 cm, with no walkways included. At sowing, two or three seeds were sown per hill, with surplus seedlings removed after emergence to maintain a uniform planting density with 20-cm plant spacing. Field management practices, including watering, fertilizing, spraying and weeding, were carried out on the basis of the local cultivation conditions.

Marker development

SSR markers were designed for gene mapping using the public database information obtained from Phytozome (<https://phytozome-next.jgi.doe.gov/>). The SSR markers were designed with BatchPrimer3 (ref. 78). These markers with polymorphisms in genotypes were used to identify all of the parental lines and populations. For SNP markers, specific primers were randomly designed on the basis of the physical position within the target interval. After PCR fragment amplification and sequencing analysis, the products exhibiting SNP differences between parental lines were identified and then used as SNP markers.

Expression pattern analysis and RNA-seq

Total RNA from sorghum root, leaf, shoot, young panicles at different developmental stages, and panicles at the anthesis stage, filling

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Author contributions

Q.X. and P.X. designed this project. P.X. and D.Z. wrote the manuscript. D.Z., P.X. and S.T. performed most of the experiments. D.Z., P.X., S.T., K.Z. and F.L. constructed multiple segregated populations and evaluated the yield traits in the field. D.Z. and P.X. collected the genotyping data for the linkage map construction. D.Z., P.X. and R.X. performed the transgenic experiments in sorghum. C.-G.D. and J.C. performed the ChIP-qPCR assay. F.L., L.K. and F. Yang performed RNA in situ hybridization. P.X., D.Z., F. Yu and C.L. analysed the data. All authors have read, discussed and contributed to the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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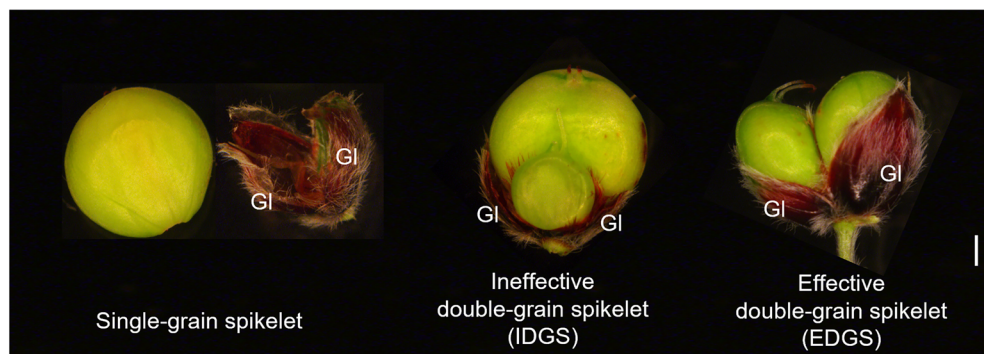
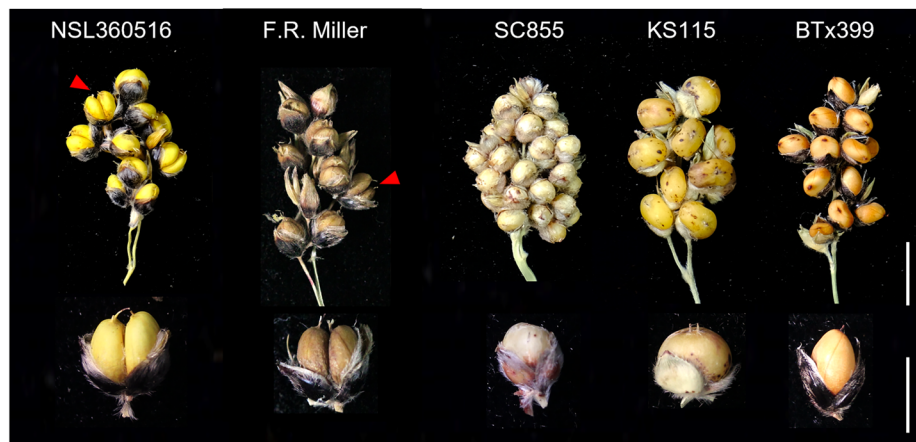
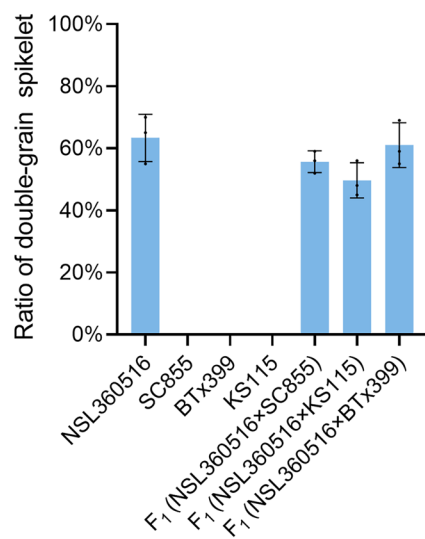
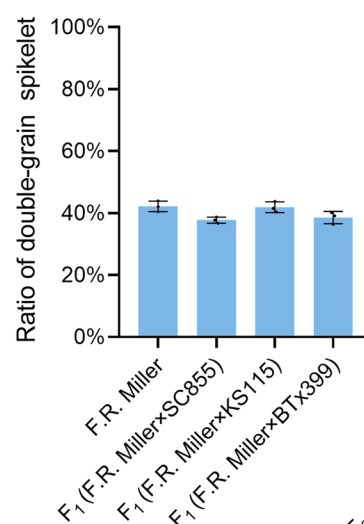
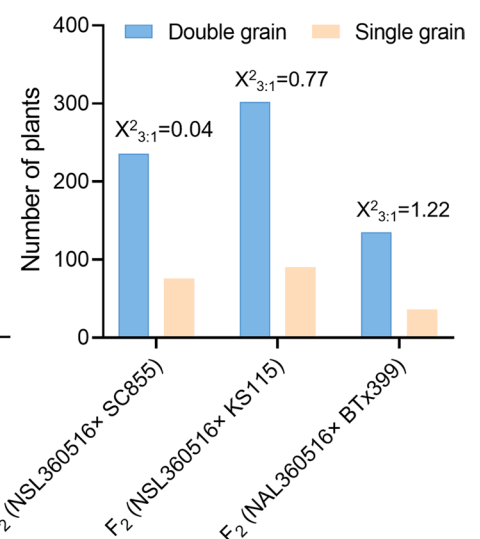
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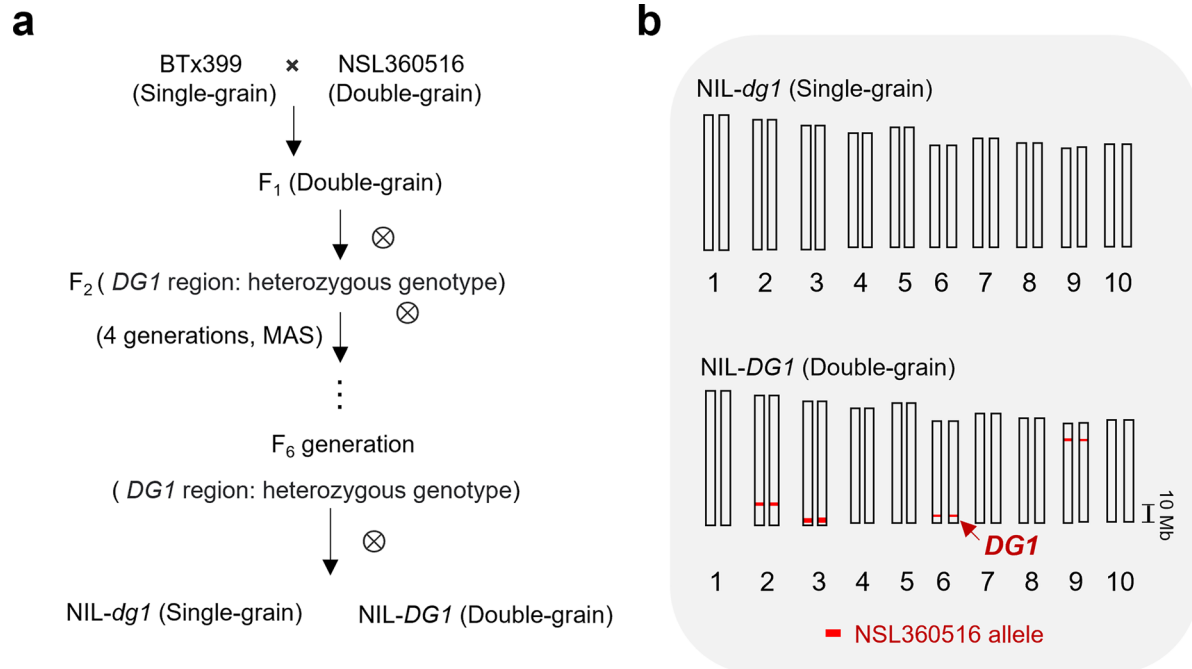
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a**b****c****d****e****Extended Data Fig. 1 | *DGI* is a single dominant locus for double-grain trait.**

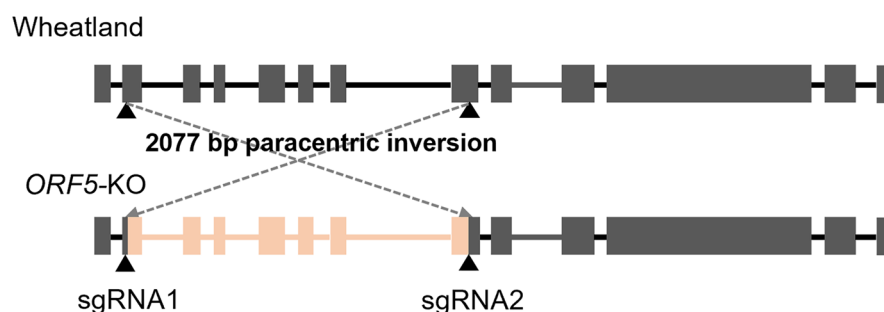
a, Seed morphology at filling stage of single-grain spikelet and IDGS and EDGS. IDGS, Ineffective double-grain spikelet. EDGS, Effective double-grain spikelet. Gl, Glume. Bar = 1 mm. **b**, Spikelet morphology of double-grain line NSL360516 and F.R. Miller, as well as three single-grain lines SC855, KS115 and BTx399. Upper and lower bar = 1 cm and 0.5 cm, respectively. **c**, Ratio of double-grain spikelet

in NSL360516, SC855, BTx399, KS115, F₁(NSL360516×SC855), F₁(NSL360516×KS115), and F₁(NSL360516×BTx399). **d**, Ratio of double-grain spikelet in F.R. Miller, F₁(F.R. Miller×SC855), F₁(F.R. Miller×KS115), and F₁(F.R. Miller×BTx399). **e**, Genetic analysis in three F₂ populations derived from F₂(NSL360516×SC855), F₂(NSL360516×KS115), and F₂(NSL360516×BTx399). Data are shown as mean ± SD, n = 3 biological replicates.



Extended Data Fig. 2 | Construction of the NIL lines. a, Through molecular markers assisted selection, the NIL lines were constructed from BTx399 and NSL360516 through self-crossing several generations. **b**, The genetic background

of NIL-*dg1* and NIL-*DG1* lines. The red boxes indicate that the genotypes are derived from NSL360516. The background in white means genotypes is the same between the two NILs.

a**b**

Wheatland	MASAQEKATSGVIRNAAAILLEMLGETCGAKKVINSELMHACAGFLVCTPCRGSTVYVFQGHSECVAAATTKKIPNSRIPNYPISLP SQ	90
ORF5-KO	MASAQEKATSGVIRNAAAILLEMLGETCGAKKVINSEMLE.....ILCQEHLCAYCQ.....	55
Wheatland	LLCQVHNITLHACKETDEIYAQMTLCPVHSETDVFFIPSLGAYTKSKHPTTEYFCNLTASDTSTHGGFVPRRAAEKLFQCLDYSMQPPN	180
ORF5-KO		55
Wheatland	QELIVRCLHDNMWTFRHIYRGQPKRHLLTTGWSLFVGAKRLKAGCSVLFIRDEKSQLLVGVRRATRQCPALSSSVLSTLSMHIGVLAAAA	270
ORF5-KO		55
Wheatland	HAASSGGSFTIYYNPTSPSPFVIPLARYNKATYLCPSVGMRFAMMFETEESIKRRCGTGTIVGISLCPMRWPNKWRNLQVEWDEHGYG	360
ORF5-KO		55
Wheatland	ERPERSLWDIETPENMVFFPSPLNSKRQCLPSYAVFGLQIGSVNMSSIPRAQGSFFGNLQQMFGSGSCLALLLNQSGCNLGSPIACQCS	450
ORF5-KO		55
Wheatland	SFSSIIQNAKHSYFPSTFGASTGSMKQETMFSNEAPQQLNALGIQKGDQVSCVQPGILCSISAREMSVKPRGPRSTDSYSSQSIDQNG	540
ORF5-KO		55
Wheatland	KGEPRTKSRSSKSSSHKTISDKSELSSLFSCICDKQRHSEPTLVCCGAEQATCGNIEDSSGAPTRGDFAGELQVLEVEQHELLSFPKL	630
ORF5-KO		55
Wheatland	ESSKSPDGGKSVSSFNQGCFSQFLEGLDWMIPPSYYQDSNGIQSATASDNIFNPFSEGI PSAMNTGMEAFQTSCLECFPNSIQEFINS	720
ORF5-KO		55
Wheatland	PDINTLTFTMSPEMQHLDAQHDGSLNLPSTNSYVQMSFSEESQSASLSGLHMEAIHINSSCSQPLATGSFDTAGTFSKLSNMKECQALPLQE	810
ORF5-KO		55
Wheatland	IHNCSMGTFSCSMDAAAEYSMDRSVKPMKPPVPTYTKVQKIGSVGRSIDVTIRFRDYHELRSIAIACMFLQGGKLEHPGGSDWKIVYVDYEN	900
ORF5-KO		55
Wheatland	DVLIVGDDPWEEFINCVRCIRILSPSEVQQMSSENGVHVLNDCIQIA	946
ORF5-KO		55

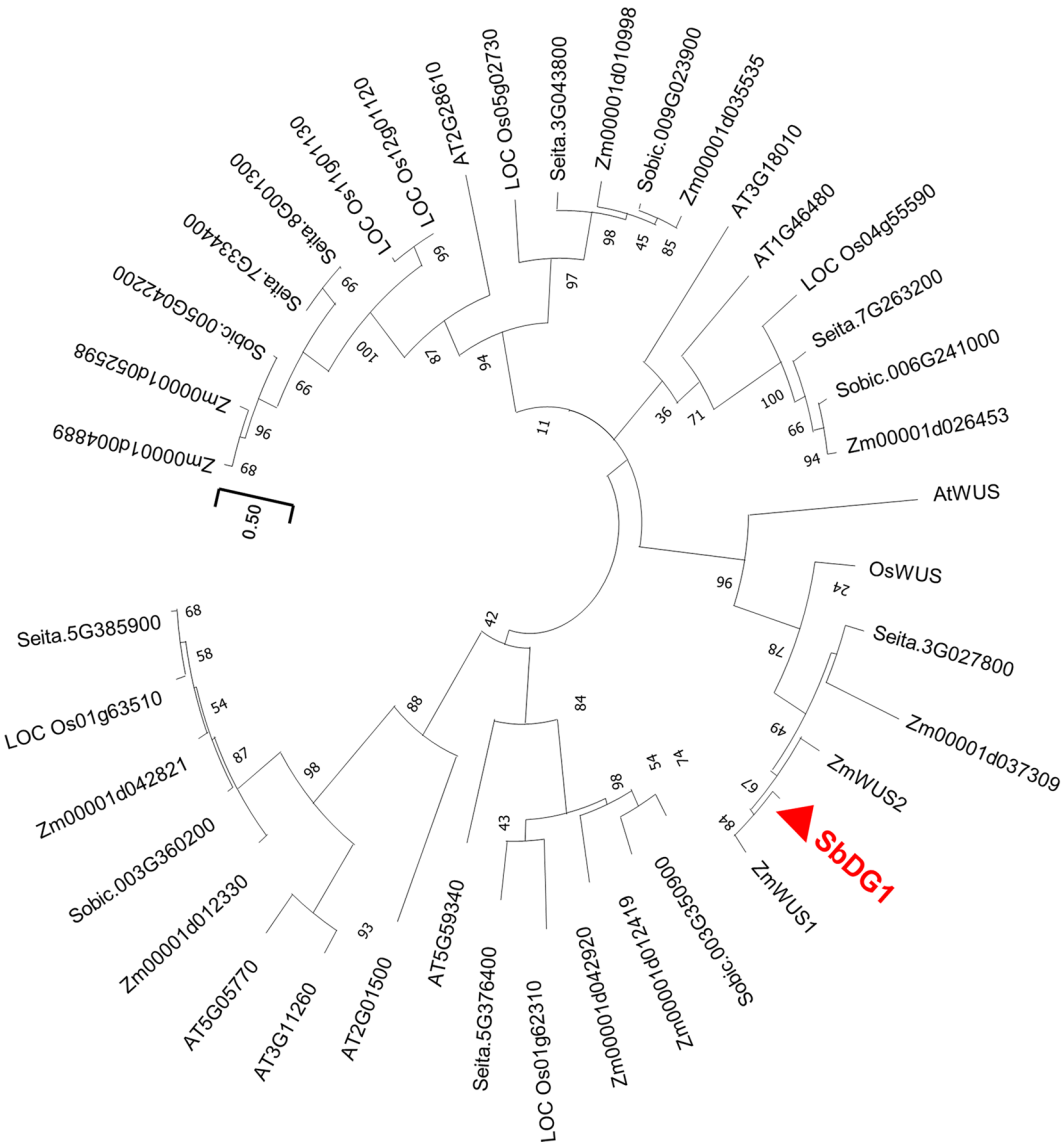
Extended Data Fig. 3 | A chromosomal paracentric inversion in *ORF5*-KO mutant. a, The chromosomal paracentric inversion fragment is 2077 bp. It occurred between sgRNA1 and sgRNA2 targets in the homozygous *ORF5*-KO line. **b**, Schematic diagram of the peptide sequence alignment of *ORF5* and *ORF5*-KO.

a

Homeobox domain

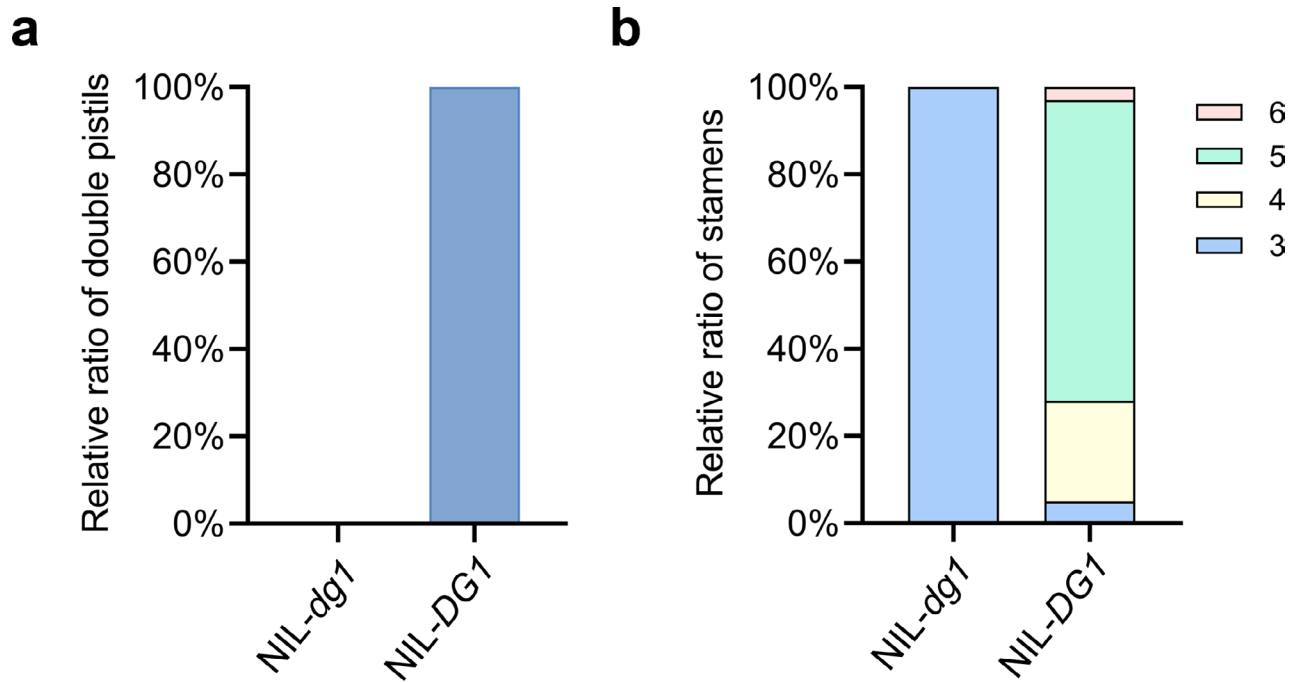
SbDG1	MAANVGAGRSVGGGGGGGAGGTGTAASGSVATTAVCRIIGSRWTPTECIRILKEFFYGCCIRSENSEICRITAMLRCHGKIEGKNVFWFCNHKAR	100
AtWUS	MEPPCHQHCHHHQADQESG...NNNNKSGSGGYT...CRQTSIRWTPTECIRILKEFFYGCCIRSENSEICRITAMLRCHGKIEGKNVFWFCNHKAR	94
ZmWUS1	MAANVGAGRSAGGGGAG...TGTGTAAGSGVS...TAVCRPSGSRWTPTECIRILKEFFYGCCIRSENSEICRITAMLRCHGKIEGKNVFWFCNHKAR	96
SbDG1	ERCKRRLTNLLVNVETAAAAGAAASTHGLVLSLSPS...GAAPPSPTLGFYAGNGGAGSTVLLDTSSDC...AAMATECTFIODYMCVMGTGSAASAAAS	194
AtWUS	ERCKRRLTNLLVNVETAAAAGAAASTHGLVLSLSPS...GAAPPSPTLGFYAGNGGAGSTVLLDTSSDC...AAMATECTFIODYMCVMGTGSAASAAAS	158
ZmWUS1	ERCKRRLTNLLVNVETAAAAGAAASTHGLVLSLSPS...GAAPPSPTLGFYAGNGGAGSTVLLDTSSDC...AAMATECTFIODYMCVMGTGSAASAAAS	185
SbDG1	PWACFSSSN.TMAAAAARAPTVTRAPETLPLFTGG...DISQPRRPRHGVFVPVAAAGAIRGGSSSSRYLFWGAAPT.TASATSIGICQHQHLLQLCEQ	290
AtWUS	PYPSENNGNLHASSGTECGVNNASNGYMSHVYGSMEQDCSMNYYN...VG...GGWANNMDHYSAPYNFDRAKFLFGLGHCHEEEECGGDAYLEHR	252
ZmWUS1	PWACSSSSSEDPMAALAPKVTTRAPETLPLFTGGGDRQPPRPRQSVF...AGAIRGGSSSSRYLFWGAAPTPTGSATSVAICQHQHLLQMQCEQ	281
SbDG1	YSFYNSA...MHGTGSCDASAASLELSLSSWCSYPAGTM...	327
AtWUS	RTLPLFP...MHGEDHINGGSGATWKYQSEVRFCALELRLN	292
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b

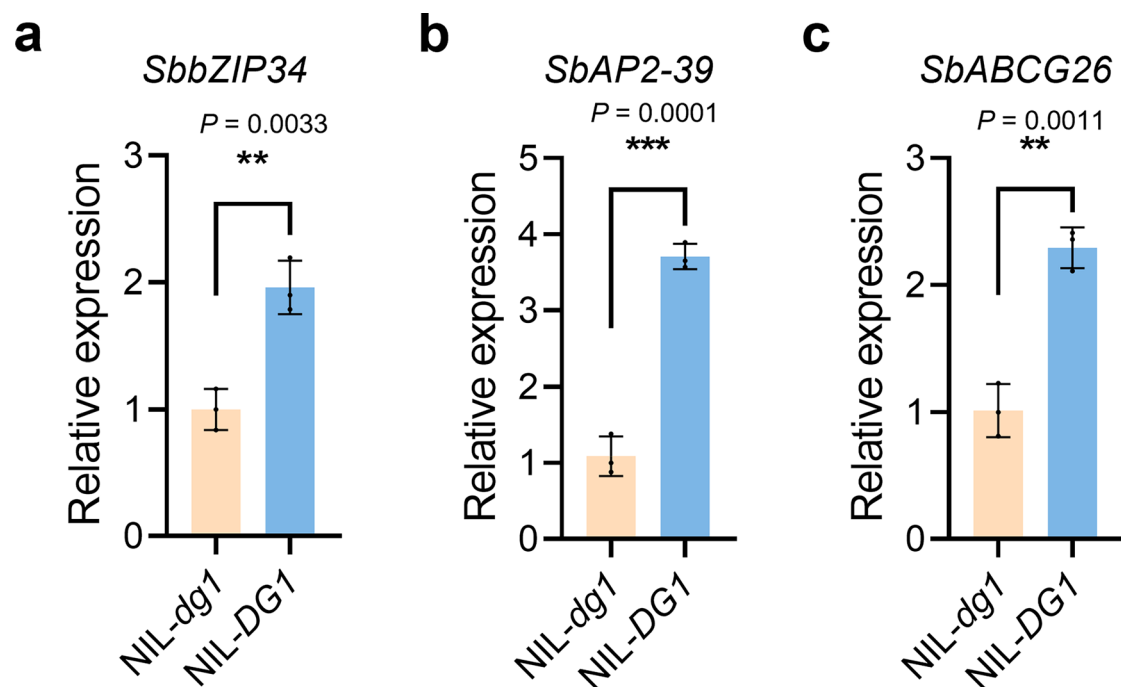


Extended Data Fig. 4 | The DG1 orthologs in crops. a, Phylogenetic analysis of DG1 and its orthologs in different species, including sorghum, maize, *Setaria italica*, rice and *Arabidopsis thaliana*. The phylogenetic tree was constructed

using the maximum likelihood method. **b**, Protein sequence alignment of SbDG1 (*Sorghum bicolor*), AtWUS (*Arabidopsis*), and ZmWUS1 (maize). The red box indicates the homeobox domain.

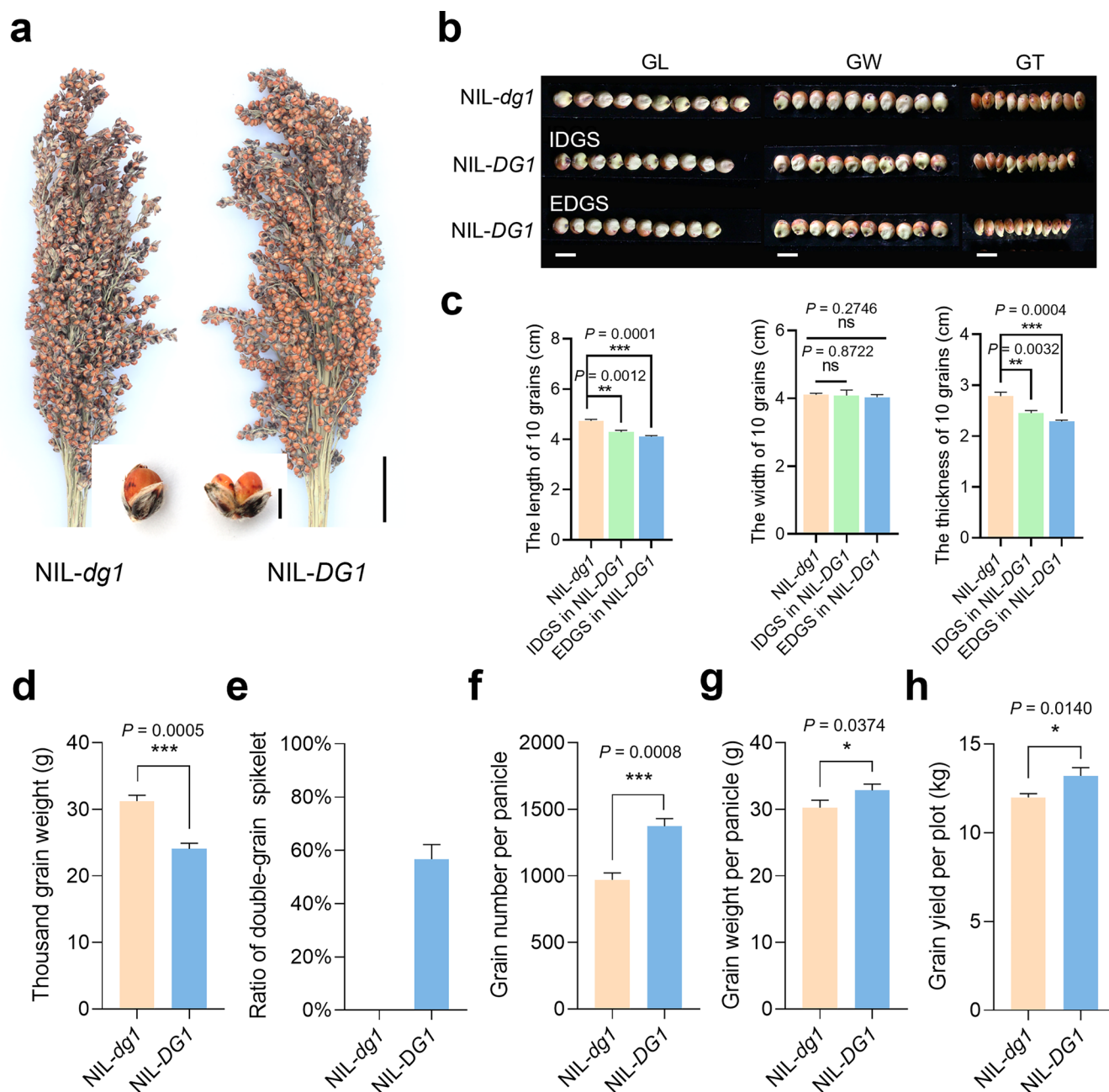


Extended Data Fig. 5 | Number of stamens and pistils in NIL-*dg1* and NIL-*DG1*. **a**, Number of double pistils in NIL-*dg1* and NIL-*DG1*. **b**, Number of stamens in NIL-*dg1* and NIL-*DG1*. For each line, 200 spikelets were examined.



Extended Data Fig. 6 | Gene expression of *SbbZIP34*, *SbAP2-39* and *SbABCG26* in NIL-*dg1* and NIL-*DG1*. **a**, Relative expression of *SbbZIP34* in the NIL-*dg1* and NIL-*DG1* at YP1 stage. **b**, Relative expression of *SbAP2-39* in the NIL-*dg1* and

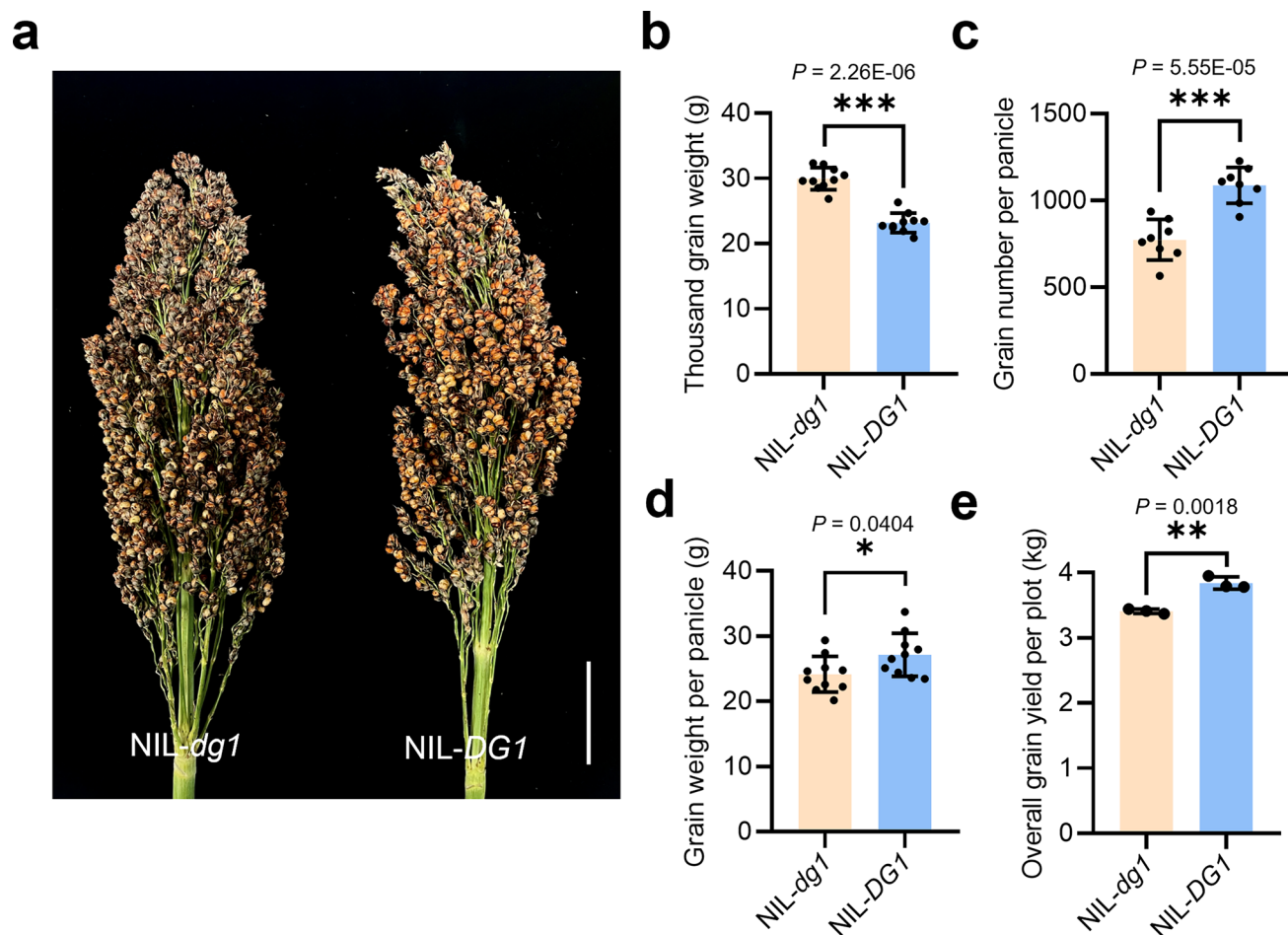
NIL-*DG1* at YP1 stage. **c**, Relative expression of *SbABCG26* in the NIL-*dg1* and NIL-*DG1* at YP1 stage. Three biological repeats were performed. *P* value was calculated by two-tailed Student's *t* test, ****P* < 0.001, ***P* < 0.01.



Extended Data Fig. 7 | *DG1* enhances grain number and grain yield in 2023.

Field yield tests were conducted in Beijing in 2023 (See Methods). **a**, Representative mature panicle and spikelet morphology of NIL-*dg1* and NIL-*DG1*. Left and right bar = 3 mm and 5 cm, respectively. **b-c**, Grain morphology and statistics of grain length (GL), grain width (GW) and grain thickness (GW) of NIL-*dg1*, IDGS and EDGS of NIL-*DG1*. *n* = 3. Bar = 5 mm. **d-h**, Statistical analysis of

thousand grain weight (*n* = 3), grain number per panicle (*n* = 3), grain weight per panicle (*n* = 3), grain yield per plot (*n* = 3; plot size: 10 meters long and 4 meters wide, with row spacing set at 50 cm and plant spacing at 20 cm) in NIL-*dg1* and NIL-*DG1*. Data are shown as mean ± SD. *P* value was determined by two-tailed Student's *t* test, *** *P* < 0.001, ** *P* < 0.01.



Extended Data Fig. 8 | *DG1* enhances grain number and grain yield in different region. Field-yield tests were conducted in ShenZhen in 2024. **a**, Representative mature panicle morphology of NIL-*dg1* and NIL-*DG1*. Bar = 5 cm. **b–e**, Statistical analysis of thousand grain weight ($n = 10$), grain number per panicle ($n = 8$), grain yield per panicle ($n = 10$), grain yield per plot ($n = 3$, plot size: 4 meters long and

3 meters wide, with row spacing set at 50 cm and plant spacing at 20 cm) in NIL-*dg1* and NIL-*DG1*. Data are shown as mean $\pm$ SD. *P* value was determined by two-tailed Student's *t* test, ****P* < 0.001, ***P* < 0.01, **P* < 0.05, ns: no significant difference.

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Sample size

For phenotype evaluation, we totally collected a diverse sorghum panel including 571 accessions preserved in our laboratory. Three F2 populations, comprising a total of 875 individuals were employed for the initial mapping of DG1. A total of 2186 individuals from two F3 and three F4 segregated populations were developed for the fine mapping of DG1, eventually pinpointed to a 49.5 Kb region. The number of individual plants was determined by the size of populations and was enough to narrow down the candidate region. Field-yield experiments were conducted at Changping Farm, Beijing, China (116.37°E, 40.10°N) during the summers of 2023 (Yield Test I) and 2024 (Yield Test II), as well as the summer of Shenzhen city, China (114.00°E, 22.80°N) (Yield Test III). Three plots for the NILs were randomly designed. For Yield tests I and II, the plot size (approximately 6,600 plants per mu) was 10 meters in length and 4 meters in width, while for Yield test III, the plot size was 4 meters long and 3 meters wide. The size of the plot is sufficient for field trial experiments. Row spacing was 50 cm with plant spacing of 20 cm, with no walkways included. At sowing, 2 to 3 seeds were sown per hill, with surplus seedlings removed after emergence to maintain a uniform planting density with 20 cm plant spacing.

Data exclusions

No data was excluded in the analysis.

Replication

Phenotype evaluation in the collected populations and field tests of NILs were performed at least two times in three different areas (Beijing, Shenzhen, and Sanya, China). All quantitative-analysis requiring experiments were performed in at least three independent biological replicates including technical replicates per experiment if feasible, such as qPCR assay. Technical replicates were averaged within each biological replicate for further analysis. All figure legends show averages and variability calculated from at least three independent experiments.

Randomization

All of the selected sorghum accessions and populations were planted under appropriate field growth conditions within the randomized block design. All independent transgenic and genome-editing sorghum plants were planted in the greenhouse randomly for phenotype evaluation. For the ratio of double-grain trait investigation, at least three randomly mature panicles of each line were selected.

Blinding

Data collection or data analysis was not blinded, but repeated by multiple independent experiments thereby reducing bias. Phenotype investigation of the collected population in the field condition were measured six times.

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Plants

Seed stocks

Novel plant genotypes

Authentication

The diverse panel of 571 sorghum accessions, sourced from the United States Department of Agriculture-Agricultural Research Service (USDA-ARS). These accessions were cultivated in both Beijing [39.5°N, 116.4°E], Shenzhen [114.00°E, 22.80°N] and Sanya [18.3°N, 108.8°E].

ORF1-overexpressing (ORF1-OE) lines were produced by transgenic method through Agrobacterium-mediated genetic transformation. Two independent positive ORF1-OE plants were obtained in the T0 generation. The expression level of ORF1 and the ratio of double-grain spikelet were investigated in the T2 generation. The ORF5-KO line was obtained by gene edited technology. Two targets (sequence: AGGTGATCAACTCCGAGCTG; GCAGCTCATGCAGCATCAAG) were designed. Then sgRNA1 and sgRNA2 with target sequences were connected to the pYLgRNA-OsU3 and pYLgRNA-OsU6a vectors, respectively. Purified gRNA expression cassettes were then inserted into the pU6-CRISPR-Cas9-MB vectors. The recombinant plasmid was introduced into the recipient sorghum line Wheatland through Agrobacterium-mediated transformation. All plasmid and vector construction methods were reported in the article (A Robust CRISPR/Cas9 System for Convenient, High-Efficiency Multiplex Genome Editing in Monocot and Dicot Plants Ma et al., 2015). After transformation, the Cas9 protein can cut the target sequences, forming double-strand breaks. Then two independent endogenous DNA repair pathways: non-homologous end joining (NHEJ) and homologous recombination (HR) were triggered, which can frequently result in small or large chromosomal changes. Six F1 plants were developed from both double-grain accessions NSL360516 and F.R. Miller, crossed with three single-grain accessions, SC855, KS115 and BTx399.